

Fabio ETERNO

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Computer engineer working as an AI engineer on end-to-end model development and deployment (NLP/LLM, classification, computer vision) for automated, higher-quality insurance claims workflows on Azure.

EDUCATION

M.Sc. Data Science and Engineering, Politecnico di Torino - Italy **2022**

- Grade=100/110, Thesis: *Differentiable Neural Architecture Search Algorithms for MLPerf Tiny benchmarks*

B.Sc. Media Communication Engineering, Politecnico di Torino - Italy **2015**

- Grade=105/110

WORK EXPERIENCE

REALE ITES SRL.....

AI Engineer **2022-Present**

Developed AI solutions in **Agile/Scrum** teams using **Azure DevOps** for **Git** version control and **CI/CD** pipelines to automate **Docker** containerization and deployment.

- **Forfait Opposition Recommendation:** Developed an automated system using **computer vision** and **deep learning** to analyze claim documentation and images, identifying circumstances warranting forfait payment opposition in inter-company claims. Deployed multi-modal pipeline with **OCR** and **CNN (PyTorch)** on **Azure**, reducing costs by **€60K per month**.
- **Mail Routing:** Developed an automated email classification system for insurance claims office using **NLP** and **NER** with **GPT-4o** via **Azure API** to extract relevant entities from incoming emails. Implemented end-to-end pipeline with preprocessing and **Random Forest** classifier to route 2,000 daily emails to appropriate claims office processes, replacing external vendor solution and saving **€100K annually**.
- **Claims Damage Assessment:** Developed an end-to-end automated damage assessment platform for auto insurance claims using **computer vision** and **machine learning**. Built **FastAPI** service to ingest accident images and meta-data into **Azure Storage**, triggering **Azure ML** batch pipeline that integrates with legacy systems for vehicle data enrichment, applies **YOLO** object detection for car identification, and uses **XGBoost** to predict damage area, severity, and repair cost estimation.
- **Expert Auditing:** Developed an automated anomaly detection platform to monitor expert assessor performance for auto insurance claims quality assurance. Built end-to-end data pipeline extracting assessment data from **Oracle** via **SQL**, orchestrated with **Azure Data Factory**, and transformed using **PySpark** on **Databricks**, storing results in **Delta Lake** tables within **Unity Catalog**. Implemented KPI computation and alert flagging to identify assessment errors and policy compliance violations, improving quality control oversight.

Data Scientist **2018-2022**

Developed machine learning solutions for industrial applications using **Python**, **PyTorch**, and **Scikit-Learn** to build predictive models and deep learning pipelines.

- **Autonomous Vehicle Image Segmentation:** Developed domain adaptation solution for autonomous vehicle perception using **adversarial neural networks** to train segmentation models on combined synthetic and real-world datasets. Implemented **deep learning** approach with **PyTorch** to improve model generalization across different driving environments, reducing dependency on expensive real-world labeled data.
- **Predictive Maintenance for Industrial Systems:** Built predictive maintenance models for industrial equipment using **regression**, **SVM** and **Random Forest** algorithms with **Scikit-Learn**. Developed end-to-end pipeline with **Python**, **Pandas**, and **NumPy** to predict equipment failures and optimize maintenance schedules, reducing downtime in manufacturing operations.

Business Intelligence Analyst **2016-2018**

Developed Business Intelligence solutions for enterprise clients, designing **ETL** pipelines and interactive dashboards using **QlikView** and **QlikSense** to support data-driven decision-making.

- **Automotive Bill of Materials Dashboard:** Developed an enterprise **Business Intelligence** dashboard for an automotive group using **QlikView** to monitor and analyze bills of materials across multiple manufacturing plants. Implemented **ETL** pipelines and optimized data models to enable tracking of component costs, inventory levels, and supply chain efficiency across the organization.

- **Treasury Management Dashboard:** Built an interactive financial dashboard for a multinational company's treasury office using **QlikSense**, consolidating data from multiple sources to provide visibility into cash flow, liquidity positions, and financial risk metrics. Designed optimized data models and implemented **ETL** processes to support strategic financial decision-making and regulatory reporting.

Digital Media Technician

2015-2016

Managed audiovisual infrastructure for high-profile sporting events including UEFA Champions League, designing equipment layouts and monitoring audio/video systems to ensure broadcast-quality production standards.

TECHNICAL SKILLS

Programming: Python, SQL, PySpark

ML/AI: PyTorch, Scikit-Learn, Deep Learning, Computer Vision, NLP, NER, YOLO, XGBoost, Random Forest, SVM

Cloud & Azure: Azure ML, Azure DevOps, Azure Storage, Azure Data Factory, Azure Pipelines, Azure OpenAI API

Data Engineering: Databricks, Delta Lake, Unity Catalog, ETL, Oracle, PySpark

API & Deployment: FastAPI, Docker, CI/CD

BI Tools: QlikView, QlikSense, PowerBi

Methodologies: Agile/Scrum, Git